Practice 2

Deadline: 2 weeks from now. Should be checked onsite (during labs).

Task 1

Design a generic type called `LuckyBox`. A lucky box can hold elements of any type. Users should be able to add elements into the box, draw one element out at random, and also look through all elements that are still inside using the enhanced `for` loop.

The following `main` method shows how the class should be used and tested. Implement the `LuckyBox` class so that the program runs correctly.

```java
public static void main(String[] args) {
    //Testing LuckyBox with different types

    // Integer box
    LuckyBox<Integer> intBox = new LuckyBox<>();
    intBox.add(1);
    intBox.add(2);
    intBox.add(3);
    intBox.add(4);
    intBox.add(5);

    System.out.println("Lucky draw (Integer): " + intBox.draw());
    System.out.println("Contents of Integer box:");
    for (Integer i : intBox) {
        System.out.println(i);
    }

    // String box
    LuckyBox<String> strBox = new LuckyBox<>();
    strBox.add("Apple");
    strBox.add("Banana");
    strBox.add("Cherry");

    System.out.println("Contents of String box:");
    for (String s : strBox) {
        System.out.println(s);
    }

    // Draw multiple times to show randomness
    // The same item should not appear again once drawn
    System.out.println("Lucky draw (String): " + strBox.draw());
    System.out.println("Lucky draw (String): " + strBox.draw());
    System.out.println("Lucky draw (String): " + strBox.draw());
    System.out.println("Lucky draw (String): " + strBox.draw());
}
```

Sample output:

```
Lucky draw (Integer): 4
Contents of Integer box:
1
2
3
5
Contents of String box:
Apple
Banana
Cherry
Lucky draw (String): Banana
Lucky draw (String): Cherry
Lucky draw (String): Apple
Lucky draw (String): null
```

Task 2

Given 1000 students, whose student IDs are **consecutive** and start with 12010001, choose an implementation to store their ages so that we can access their ages quickly using their student IDs.

Implementation 1:

Probably the most intuitive implementation is to use `HashMap<Integer, Integer>`, where key is student ID and value is age.

Implementation 2:

Another choice is to use `ArrayList` to store ages; when accessing the age, use `index = student ID - 12010001`.

Implementation 3:

The third choice is to use `LinkedList` to store ages; when accessing the age, use `index = student ID - 12010001`.

Implementation 4:

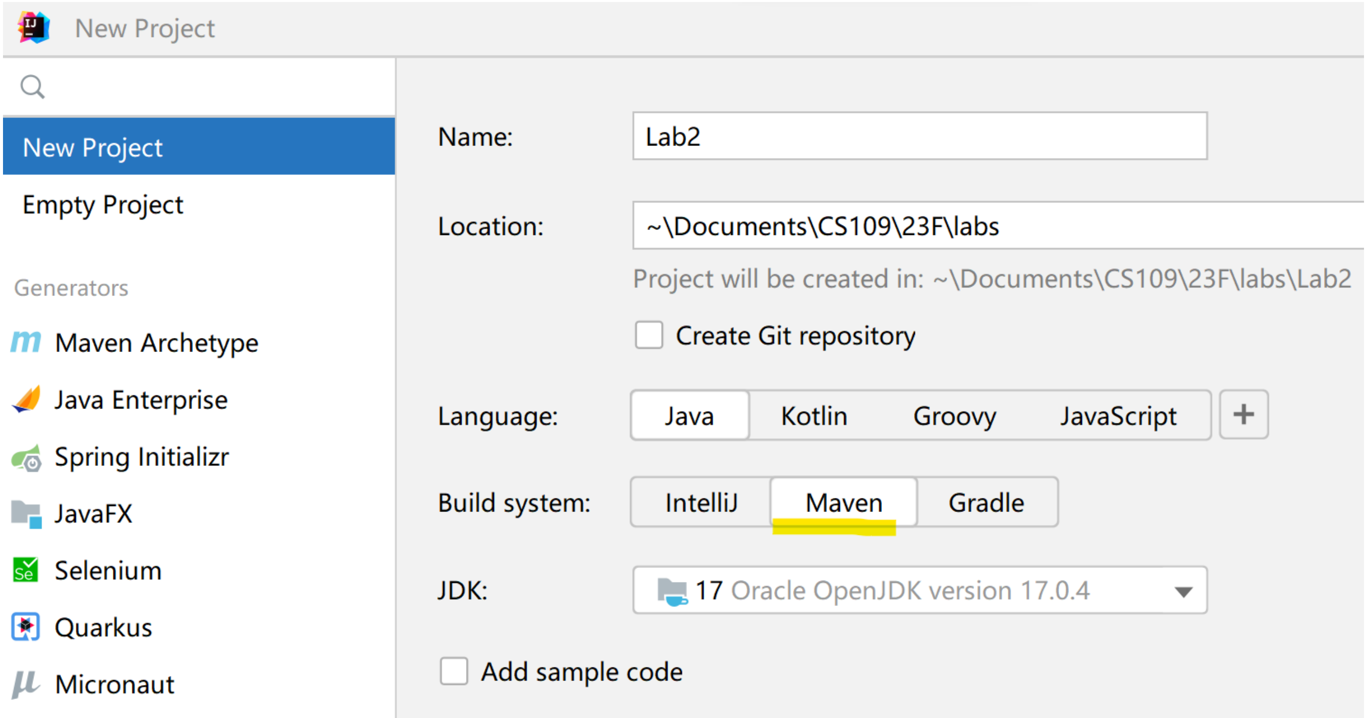
We can also use `int[]` to store ages; when accessing the age, use `index = student ID - 12010001`.

Installing JMH

We'll use the `JMH` (Java Microbenchmark Harness) library to profile different implementations and see which implementation is faster.

Create a new Maven project in IntelliJ (File->New->Project->Maven). (In IntelliJ Ultimate, if you already have an existing project that's not a Maven project, right-click the project and "Add Framework Support", then click

Maven. However, this feature is not available in IntelliJ Community).



There will be a `pom.xml` in a Maven project. Open its `pom.xml` file and add the following dependencies.

```
<dependencies>
    <dependency>
        <groupId>org.openjdk.jmh</groupId>
        <artifactId>jmh-generator-annprocess</artifactId>
        <version>1.35</version>
    </dependency>
    <dependency>
        <groupId>org.openjdk.jmh</groupId>
        <artifactId>jmh-core</artifactId>
        <version>1.35</version>
    </dependency>
</dependencies>
```

Right-click `pom.xml`, click "Maven->Reload project", and maven will automatically download these jars for you if you haven't done so.

Running JMH

Now, we can use JMH in our project. In `src/main/java`, create a package `practice.lab2` and put `Practice2.java` in it. We've already implemented `HashMap<Integer, Integer>` (Implementation 1). Specifically, initializing the HashMap is done in `MyState.setUp` and getting the age by student ID is done in `testintmap()`.

In this practice, please implement the problem also using `ArrayList` (Implementation 2), `LinkedList` (Implementation 3) and `int[]` (Implementation 4) by filling in the `TODO` in `Practice2.java`. Specifically, you should also perform initialization in `MyState.setUp` and element accessing in `testarraylist`, `testlinkedlist` and `testarray`.

Finally, executing the `main` method.

Sample output

If you have done everything correctly, you should see that JMH is calculating the running time of `test*` methods (which may take a while) and finally output the following benchmark results (results may vary on different machines). You should submit this output to us.

```
Benchmark                      Mode  Cnt    Score      Error  Units
Practice2Answer.testarray      avgt    3    1.050 ±    0.069  ns/op
Practice2Answer.testarraylist  avgt    3    2.024 ±    2.246  ns/op
Practice2Answer.testintmap     avgt    3    4.275 ±   10.259  ns/op
Practice2Answer.testlinkedlist avgt    3  157.808 ± 2118.662  ns/op
```

Evaluation

The practice will be checked by teachers or SAs. What will be tested:

1. That you understand every line of your own code, not just copy from somewhere
2. That your program compiles correctly (javac)
3. Correctness of the program logic
4. That the result is obtained in a reasonable time

Late submissions after the deadline will incur a 20% penalty, meaning that you can only get 80% of this practice's score.